

Improving the stability of multiple-relaxation lattice Boltzmann methods with central moments

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ABSTRACT

In this work we seek lattice Boltzmann methods (LBM) with improved stability that retain accuracy. Using von Neumann stability analysis to extract dispersion and diffusion errors, we compare the D2Q9 lattice scheme with three collision operators: the single relaxation time (BGK), the multiple-relaxation time with raw moments (MRT-RM) and with central moments (MRT-CM). First, we observe that the MRT-CM shows favorable properties when compared to the other two schemes. This method provides low errors, enhanced stability and enables the modification of free parameters. Second, we optimize these free parameters to increase dissipation only for high under-resolved wavenumbers, leaving low wavenumbers (well resolved scales) unchanged. In particular, we show that the optimized MRT-CM can cope with lower viscosities, higher velocities and coarser meshes than their predecessors. Finally, the optimized MRT-CM is tested for a shear layers flow to illustrate the enhanced stability and accuracy of the proposed technique.

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1. Introduction

Over the last two decades, the Lattice Boltzmann method (LBM) has become a popular alternative to traditional discretization techniques of the Navier–Stokes (NS) equations to simulate fluid flows. The underlying idea of the LBM is to construct kinetic models that recover the NS equations in the macroscopic limit (Chapman–Enskog expansion [1]). Large part of the responsibility of modeling the physics correctly falls on the collision operator, which has an strong effect in the numerical stability of the scheme [2,3].

The most popular collision operator, the single-relaxation time, is based on the BGK [4] approximation. In this model a unique relaxation time is considered for all the probability distribution functions. As a result of its simplicity, it has severe stability limitations [5].

The multiple-relaxation time with raw moments (MRT-RM) collision operator [6] was introduced in an attempt to improve the BGK stability limitations. This operator performs the collision in momentum space instead of the usual velocity space used by the BGK. Furthermore, the MRT-RM enables different relaxation times for each probability distribution function. The increased complexity with respect to BGK results in improved numerical stability [7].

Despite the enhancement in the stability of the MRT-RM with respect to the BGK collision operator, the MRT-RM may show instabilities for small fluid viscosities [8]. Consequently, the MRT with central moments (MRT-CM) collision operator was introduced [9]. Unlike the MRT-RM, where the moments are calculated in the global reference frame; the MRT-CM is calculated with the local macroscopic velocity. Using this formulation for the collision operator in the space of moments, it is possible to obtain a higher numerical stability.

von Neumann stability analysis [10] enables the quantification of numerical errors in the numerical schemes. Sterling and Chen [11] were the first to apply this type of stability analysis to the Lattice Boltzmann methods. Then, Lallemand and Luo [7] used this analysis to compare the enhanced stability of the MRT-RM over the BGK approach. Subsequently, Siebert et al. [12] included high order terms into the equilibrium distribution of the D2Q9 model to enhance further the linear stability of the scheme (as shown depicted Fig. 2 in [12]). More recently, Malaspinas [13] proposed a new version of the BGK with improved stability and based on recursive relations and regularization for the LB posed as Hermite series, which has been subsequently validated by Mattila et al. [14] and Coreixas et al. [15]. Besides studying stability, this technique has been successfully used to provide insight into dispersive and dissipative errors in LBM, in the past. Marié et al. [16] compared BGK and MRT-RM collision operators. Dubois et al. [17] studied the numerical stability of the relative veloc-

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ity (MRT–RM and MRT–CM) D2Q9 schemes with two different set of moments, proposed by Lallemand and Luo [7] and Geier et al. [8]. They concluded that MRT–CM with Geier's conservation law [8] had better stability properties.

As it was mentioned before, both the MRT–RM and the MRT–CM relax each hydrodynamic moment with a different relaxation time but all combinations lead to the same macroscopic state through the Chapman–Enskog expansion [1]. The relaxation times that are not fixed by the physics become free parameters that can be optimized to enhance particular numerical features. Lallemand and Luo [7] optimized these parameters (for the D2Q9 lattice scheme) maximizing the Galilean invariance of the scheme, while reducing numerical errors (i.e. dispersion and dissipation). Similarly, Xu and Sagaut [18], and Xu et al. [19] proposed an optimization of the free parameters to minimize dispersion/dissipation errors for the MRT–RM in D2Q9 scheme.

In this work, the von Neumann stability analysis has been applied for three different collision operators: BGK, MRT–RM and MRT–CM. The bulk and kinematic viscosity has been set to the same value establishing a fair comparison between the BGK and the MRT, to clarify how the MRT free relaxation times affect dispersion and dissipation errors. Moreover, to improve the MRT–CM, a different optimization strategy is proposed. Instead of minimizing dissipation errors for all wavenumbers, we propose to maintain numerical dissipation for well resolved wavenumbers whilst increasing dissipation for under-resolved wavenumbers. The optimization is inspired in the rule of k–1% dispersion-error presented by Moura et al. [20] in the context of high order numerical methods. They suggested that waves are only accurately resolved if the dispersion error (difference between theoretical and numerical) is below 1%. The wavenumber at which the error becomes 1% was named “k–1% dispersion-error” and lead to the “1% rule”. Following this rule, all waves above the k of 1% should be dissipated since these will be poorly resolved and may pollute the solution. We follow the idea of damping under-resolved waves and apply it to the LBM for the first time.

The remaining of this text is organized as follows. First, in Section 2, the three lattice Boltzmann formulations BGK, MRT–RM and MRT–CM are briefly introduced. In Section 3, the numerical stability of the three collision operators is compared by means of the von Neumann analysis. The effect of the free parameters of the MRT–RM and MRT–CM on hydrodynamic (shear and acoustic modes) and kinetic modes is analyzed. Then, in Section 4 the optimization process is detailed for the MRT–CM approach. Finally, in Section 5 the advantages of our selection of relaxation parameters are tested for a double shear periodic layer.

2. Lattice Boltzmann methods

2.1. Lattice Boltzmann equation

The continuous Boltzmann equation reads:

$$\frac{\partial f}{\partial t} + \xi \cdot \nabla f = \Omega_f, \quad (1)$$

where the probability distribution function (PDF) $f(\mathbf{x}, \xi, t)$, is defined such that $f(\mathbf{x}, \xi, t)d\mathbf{x}d\xi$ represents the volume of particles in the phase space element $d\mathbf{x}d\xi$ at a given time t , see Liboff [21] for details. And Ω_f is the collision operator.

The LBM is a numerical technique that provides numerical solutions of Eq. (1). The discrete set of velocities is a vector of Q components represented by the symbol $\mathbf{e}_i$. As a consequence, the probability distribution function is not continuous and the Q discrete probability distribution values $f_i(\mathbf{x}, \mathbf{e}_i, t)$ are stored at each lattice node for each time step, Δt . The discrete Boltzmann equation,

Eq. (1) becomes:

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = \Delta t \cdot \Omega_i(\mathbf{x}, t), \quad i = 0, \dots, Q - 1, \quad (2)$$

At each time step, the information stored in the discrete probability distribution functions is streamed through the lattice and collided. The discrete collision operator Ω_i is the responsible of computing the post-collision state conserving mass and linear momentum.

Finally, it should be mentioned that, $\mathbf{e}_i$ is scaled by the particle or propagation speed c , which in the classic approach is related to the lattice size and the time step; i.e. $c = \frac{\Delta \mathbf{x}}{\Delta t} = 1$, such that $\Delta \mathbf{x} = \Delta t$.

2.2. Collision operator

The collision operator is of critical importance in the LBM method, as it is responsible of the proper modelization of the physics. The high cost of its evaluation precluded its use until Higuera and Jimenez [3] simplified this operator by performing linearization (under the assumption that the discrete probability distribution function (PDF) after the collision, f_i , is close to its equilibrium state). Expanding the discrete collision operator, Ω_i , around the equilibrium state of the discrete probability distribution function, f_i^{eq} , leads to a linearized operator:

$$\Omega_j = K_{ji}(f_i^{\text{eq}} - f_i), \quad (3)$$

where K_{ji} is known as collision matrix. In this work the convention of assumed summation over repeated indices is followed.

The discrete local equilibrium function f_i^{eq} is computed as a second order Taylor expansion of the Maxwell–Boltzmann distribution,

$$f_i^{\text{eq}} = \rho w_i \left(1 + \frac{\mathbf{e}_i \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right), \quad (4)$$

where ρ is the density, w_i are weighting constants built to preserve isotropy, $\mathbf{u}$ is the macroscopic velocity and c_s is the speed of sound. The particular values of the weighting constants w_i , depend on the discrete set of velocities [22].

The collision matrix describes how the PDFs relax towards the equilibrium state. Three approaches are considered in this work and are detailed in the following sections.

2.2.1. Single-relaxation time

The single-relaxation time based on the Bhatnagar–Gross–Krook (or BGK) [4] approximation, where $K_{ji} = (1/\tau)\delta_{ij}$ (assuming Kroencker delta notation), is the most popular approach. In this case the collision operator simplifies to:

$$\Omega_i^{\text{BGK}} = \frac{1}{\tau}(f_i^{\text{eq}} - f_i). \quad (5)$$

Only one relaxation time for all PDFs, τ , is considered. The Chapman–Enskog expansion applied to the classical LBM equation (Eq. (2)), with Eq. (5) as the discrete collision operator, establishes the relation between the relaxation time, the kinematic, ν , and bulk viscosities, μ_ν , of the macroscopic fluid [23]:

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right), \quad \mu_\nu = \frac{2}{D} \nu. \quad (6)$$

where D is the spatial dimension.

The BGK collision operator, is still common but has several shortcomings. For example, the kinematic viscosity is conditioned by the relaxation parameter, τ , (see Eq. (6)) which causes numerical instabilities for values near $\tau = 0.5$ (with $\Delta t = 1$), therefore complicating the simulation of high Reynolds number flows [5]. It should also be noticed that the bulk viscosity, μ_ν , cannot be freely chosen because it is constrained by the kinematic viscosity, ν .

2.2.2. Multiple-relaxation time

The multiple-relaxation time with raw moments (MRT–RM) was developed in an attempt to overcome some of the limitations of the BGK [6]. This method performs the collision process in momentum space instead of in velocity space. First developments were extended for the D2Q9 scheme [7] and for the D3Q19 scheme [24]. The increased flexibility in the selection of the relaxation parameters with respect to the BGK approach resulted in an enhanced stability when compared with the BGK [7]. Despite its increased stability, the MRT–RM still shows instabilities for small viscosities [8], due to the lack of Galilean invariance, which can result in negative viscosities. The multiple-relaxation time with central moments (MRT–CM) improved some of the shortcomings of the MRT–RM by calculating the moments with respect to a local velocity [8,9]. By shifting the discrete velocities with the local macroscopic velocity, it is possible to obtain a higher degree of Galilean invariance compared with the MRT–RM approach.

Recently, Dubois et al. [25] have demonstrated that both MRT–RM and MRT–CM are particular cases of relative velocity schemes, which enabled to prove the relation between both collision operators. In the following sections, Dubois' notation will be used to unify the formulation of MRT–RM and MRT–CM. In both models the relaxation matrix, will be computed as a product of three matrices $K_{ji} = M_{jk}^{-1} S_{kl} M_{li}$. The matrix M accounts for the definition of the moments while the matrix S is defined as a diagonal matrix:

$$S = \text{diag}(\mathbf{s}), \quad (7)$$

where $\mathbf{s}$ is a vector with the relaxation times of the different moments.

MRT with Central Moments:

The MRT–CM collision operator is obtained defining the collision matrix as $K_{ji} = M_{jk}(\mathbf{u})^{-1} S_{kl} M_{li}(\mathbf{u})$:

$$\Omega_j^{\text{MRT--CM}} = M_{jk}(\mathbf{u})^{-1} S_{kl} M_{li}(\mathbf{u}) (f_i^{\text{eq}} - f_i), \quad (8)$$

This notation allows working with the equilibrium PDF, f_i^{eq} defined in the BGK model. Note that the BGK collision operator can be recovered by setting all the diagonal elements, $\text{diag}(\mathbf{s})$, to $1/\tau$.

MRT with Raw Moments:

Similarly to the MRT–CM scheme, the raw moments version sets $\mathbf{u} = 0$ and the collision matrix becomes $K_{ji} = M_{jk}(0)^{-1} S_{kl} M_{li}(0)$. The collision operator is:

$$\Omega_j^{\text{MRT--RM}} = M_{jk}(0)^{-1} S_{kl} M_{li}(0) (f_i^{\text{eq}} - f_i). \quad (9)$$

The formulation introduced up to this point is general for any lattice distribution and spatial dimensions. Without loss of generality in the methodology, the stability study and optimization of the following sections will be performed in two dimensions and with 9 velocities. This is known as the D2Q9 lattice scheme, where x and y components of the $\mathbf{e}_i$ microscopic velocity vector are $e_x = (0, 1, 0, -1, 0, 1, -1, -1, 1)$ and $e_y = (0, 0, 1, 0, -1, 1, 1, -1, -1)$ (assuming that the propagation speed is $c = 1$). The speed of sound, c_s , is constrained by the lattice scheme to $c_s = 1/\sqrt{3}$.

The $M(\mathbf{u})$ and $M(0)$ matrices are detailed in Appendix A. Finally, as it was explained before, S is a diagonal matrix. The entries of this diagonal matrix account for the relaxation times (s_i) of the different moments. In particular, in the D2Q9 scheme this reads:

$$S = \text{diag}(s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8). \quad (10)$$

As can be seen in Appendix A, in the case of MRT collision model the hydrodynamic quantities (density and momentum) can be imposed with any value of s_0, s_1 and s_2 , provided $\rho = \rho^{\text{eq}}$ and $\mathbf{u} = \mathbf{u}^{\text{eq}}$. In what follows, they are set to zero.

As for the BGK collision operator, the Chapman–Enskog expansion establishes the relation between the relaxation times and the

kinematic and bulk viscosities of the macroscopic fluid [7]. In the MRT approach the kinematic viscosity ν and the bulk viscosity μ_v are related to the relaxations parameters (s_3, s_4, s_5) associated to the second order moments. As a result, this approach removes some of the constraints of the BGK collision operator. In particular, the viscosities read:

$$\nu = c_s^2 \left(\frac{1}{s_4} - \frac{\Delta t}{2} \right) = c_s^2 \left(\frac{1}{s_5} - \frac{\Delta t}{2} \right), \quad \mu_v = \frac{2}{D} c_s^2 \left(\frac{1}{s_3} - \frac{\Delta t}{2} \right). \quad (11)$$

It is important to notice that, having identified the relations between the kinematic and bulk viscosities with the relaxation times, the higher order moments relaxation times, s_6, s_7 and s_8 , remain as free parameters. Finally, it should be mentioned that the relaxation values, s_i , related with the viscosities must lie between 0 and 2 since the kinematic and bulk viscosity cannot be negative (see Eq. (11) with $\Delta t = 1$).

3. Numerical stability

3.1. von Neumann stability analysis

The von Neumann (VN) stability analysis [10] provides information about the stability of a numerical scheme by studying how the different Fourier modes propagate through the scheme. Consequently, it provides insights into the dispersion and dissipation errors of the underlining numerics.

Applying the VN analysis to the LBM with the different collision operators (BGK, MRT–RM and MRT–CM), amplification matrices G_{ij} are obtained. Details on the derivation of the amplification matrices are provided in Appendix B.

The amplification matrices, G_{ij} , for the different collision operators analyzed are:

$$\begin{aligned} G_{ij}^{\text{BGK}} &= A_{ik}^{-1} \left[\delta_{kj} + \frac{\Delta t}{\tau} N_{kj} \right], \\ G_{ij}^{\text{MRT--RM}} &= A_{ik}^{-1} \left[\delta_{kj} + \Delta t M_{kl}(0)^{-1} S_{lm} M_{mn}(0) N_{nj} \right], \\ G_{ij}^{\text{MRT--CM}} &= A_{ik}^{-1} \left[\delta_{kj} + \Delta t M_{kl}(\mathbf{u})^{-1} S_{lm} M_{mn}(\mathbf{u}) N_{nj} \right], \end{aligned} \quad (12)$$

with $\begin{cases} A_{ik} = \delta_{ik} e^{i\mathbf{k} \Delta t \mathbf{e}_i}, \\ N_{nj} = \left. \frac{\partial f_j^{\text{eq}}}{\partial f_j} \right|_{\bar{f}_n} - \delta_{nj}. \end{cases}$

An eigenvalue analysis of the amplification matrices, G_{ij} , provides information of the growth rate and wavenumber, $k = \{k_x, k_y\}$, of the different Fourier modes that represent perturbations around an equilibrium solution, $\bar{f}_i$. From the eigenvalue problem, $e^{-i\omega \Delta t} F_i = G_{ij} F_j$, for F_i the eigenvectors with complex eigenvalues defined as:

$$\lambda = e^{-i\omega \Delta t}, \quad (13)$$

where i is the imaginary unit and ω is the frequency in time. Then, the transport coefficients can be obtained by logarithmic transformation into Eq. (13):

$$\omega = \frac{i}{\Delta t} \ln(\lambda), \quad (14)$$

and they are directly associated with the numerical errors. The von Neumann stability condition $|\lambda| \leq 1$ can be deduced from Eq. 13, meaning that a numerical scheme is considered stable if perturbations are not exponentially amplified. Additional, information about the dispersion and dissipation of the numerical scheme can be extracted from the transport coefficients. In particular, dispersion and dissipation are related with the real, $\text{Re}(\omega)$, and imaginary, $\text{Im}(\omega)$, parts of the transport coefficient, respectively.

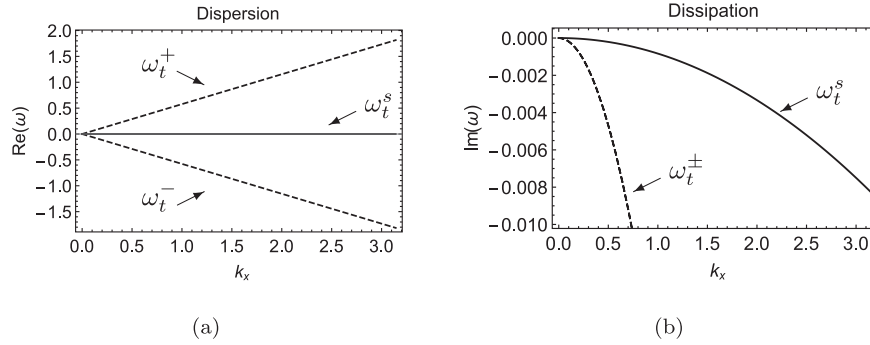


Fig. 1. Dispersion, $\text{Re}(\omega)$, (a) and dissipation, $\text{Im}(\omega)$, (b) of the theoretical modes (shear mode, ω_t^s , —, and acoustic modes, $\omega_t^\pm$, - -) at $\text{Ma} = 0$ with $\nu = 10^{-3} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.988$) and $\mu_\nu = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$).

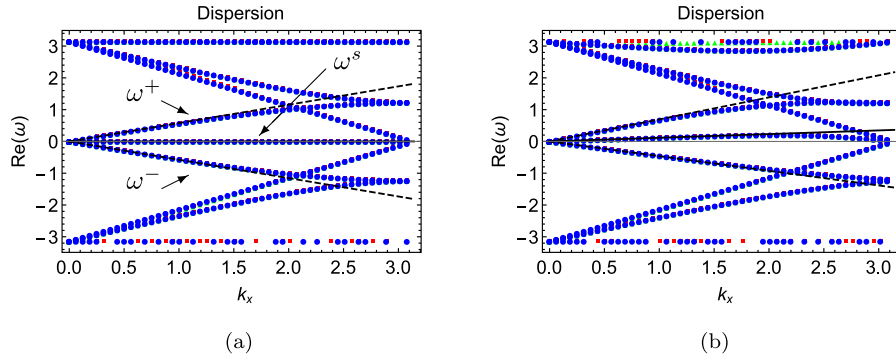


Fig. 2. Dispersion for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators and theoretical modes (shear mode, —, and acoustic modes, - -) at $\text{Ma} = 0$ (a) and $\text{Ma} = 0.2$ (b) with $\nu = \mu_\nu = 10^{-3} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.988$).

3.2. Numerical analysis for BGK, MRT-RM and MRT-CM with D2Q9

The aim of the numerical analysis performed in this section is to study the stability properties of the LBM with BGK, MRT-RM and MRT-CM. To this end, the VN analysis is applied to these schemes to obtain a numerical behavior which is compared to the theoretical hydrodynamic modes.

3.2.1. Theoretical modes

The theoretical modes are obtained through an analytic linearization of the Navier–Stokes (NS) equations [26]. The resulting hydrodynamic modes are known as shear mode (ω_t^s) and acoustic modes ($\omega_t^\pm$). The former is related to the kinematic viscosity and the latter are related to both kinematic and bulk viscosities. The analytical expressions for the theoretical hydrodynamic modes are:

$$\begin{aligned} \omega_t^s &= |\mathbf{k}| |\mathbf{u}| \cos \phi - i |\mathbf{k}|^2 \nu, \\ \omega_t^\pm &= |\mathbf{k}| (|\mathbf{u}| \cos \phi \pm c_s) - i |\mathbf{k}|^2 \left(\frac{(D-1)}{D} \nu + \frac{\mu_\nu}{2} \right), \end{aligned} \quad (15)$$

where ω are complex numbers, ϕ is the angle between the perturbation vector (represented through its wavenumber $\mathbf{k} = \{k_x, 0\}$) and the streaming velocity direction $\mathbf{u}$. In what follows, ϕ is considered zero (i.e. perturbation aligned with the streaming velocity). Eq. (15) shows that the theoretical dispersion (real part) is independent of the viscosities which only affect dissipation (imaginary part). Fig. 1 depicts dispersion (real part of ω , $\text{Re}(\omega)$) and dissipation (imaginary part of ω , $\text{Im}(\omega)$) behaviour for a range of wavenumbers $[0, \pi]$, following Eq. (15). Finally, note that this error analysis uses a linearized solution for the NS equations and not a linear advection-diffusion equation, as typically for VN analysis [20,27].

3.2.2. Numerical modes

Having derived the theoretical expression for the three collision operators (BGK, MRT-RM, MRT-CM), we can compare them with the modes issued from the numerical discretization. To this end, the von Neumann stability analysis is applied to the LBM with the three collision operators. It should be mentioned that the D2Q9 discretization results in nine numerical modes, where three of them are the hydrodynamic modes: one shear mode (ω^s) and two acoustic modes ($\omega^\pm$). The rest of them corresponds to kinetic modes.

Recall that the bulk viscosity in the BGK approach is linked to the kinematic viscosity $\mu_\nu^{\text{BGK}} = \frac{2}{D} \nu$, while in the MRT approaches there is no link between them since $\mu_\nu^{\text{MRT}} = \frac{2c_s^2}{D} (\frac{1}{s_3} - \frac{1}{2})$. This impacts in the diffusive behavior of the schemes. It follows that the shear and acoustic modes have the same behavior in the BGK scheme, but can be different in the MRT models (as observed for the theoretical modes (Eq. (15))).

In this section the effect of the Mach number, kinematic and bulk viscosities in the numerical modes is studied. The Mach number is defined as $\text{Ma} = U_0/c_s$, where U_0 is the reference velocity. Two different Ma are retained, $\text{Ma} = 0$ and $\text{Ma} = 0.2$. To analyze the effect of viscosity, this section has been divided in three subsections. First, we fix the bulk and kinematic viscosities to the same value and compare the three collision operators. Second, the instabilities in the MRT-RM scheme at low viscosity condition are illustrated. Third, we show the effect of varying the bulk viscosity on the acoustic modes.

Equal kinematic and bulk viscosities:

The numerical analysis presented in this section is based on the work of Marié et al. [16] where the BGK and MRT-RM collision operators were compared using the relaxation times proposed by Lallemand and Luo [7] with D2Q9 as lattice scheme. In addition, to

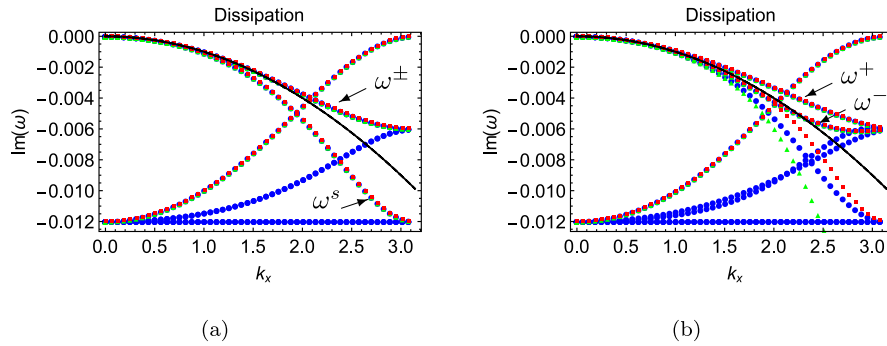


Fig. 3. Dissipation for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators and theoretical modes (shear mode, —, and acoustic modes, —) at $\text{Ma} = 0$ (a) and $\text{Ma} = 0.2$ (b) with $\nu = \mu_\nu = 10^{-3} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.988$) (Note that the theoretical modes are overlapped).

perform a comparison between BGK and MRT collision operators, we fix the bulk viscosity in the BGK and the MRT approaches to the same value. In particular, both kinematic and bulk viscosities are fixed to $10^{-3} \frac{\text{kg}}{\text{ms}}$. That means a value of $\tau = 0.503$ for the BGK collision operator and values $s_3 = s_4 = s_5 = 1.988$ for the MRT collision operators. As far as the free parameters of the MRT collision operators (s_6 , s_7 and s_8), these are fixed to the values proposed by Lallemand and Luo [7] (but they will be varied in subsequent sections). To sum up, the diagonal matrix of the collision matrix of the MRT collision operators considered in Figs. 2 and 3 reads:

$$S = \text{diag}(0, 0, 0, 1.988, 1.988, 1.988, 1.9, 1.9, 1.54). \quad (16)$$

Fig. 2 shows the dispersion of the three collision operators compared with the theoretical modes for two different Mach numbers (0 and 0.2). Notice that the three approaches have comparable dispersion errors. The dispersion error at low wavenumbers is negligible, but for large wavenumbers we observe the development of non-negligible dispersion errors. The asymmetry, with respect to the horizontal axis, shown in Fig. 2(b) is a result of constant streaming velocity (for $\text{Ma} = 0.2$).

Fig. 3 shows dissipation where the theoretical modes (shear and acoustic) overlap, which is due the kinematic and bulk viscosities having the same value (see Eq. (15)). Also, one can observe in Fig. 3(a) how the shear and acoustic modes remain unaltered for the three approaches. Fig. 3(b) shows that the acoustic modes start to separate due to the delayed wave produced by the streaming velocity. This behavior is observed in both acoustic branches. Note that the MRT-CM presents a higher dissipation rate at high wavenumbers in the shear mode. This will prove advantageous to dissipate under-resolved waves (see following sections).

One of the main properties of the MRT collision operators is that it presents higher dissipation than the BGK method for the kinetic modes. Although this could seem a disadvantage (the dissipation is overestimated) it has an stabilization effect on the solution. Whether this affects positively or negatively to the accuracy of the numerical scheme depends on the dispersion properties of the scheme at those high wavenumbers. If these wavenumbers are not propagated at the right speed, they are only introducing and propagating numerical errors into the solution and could be damped. This idea was suggested by Moura et al. [20].

Kinematic viscosity instabilities:

A well known advantage of MRT-CM with respect to MRT-RM is its higher stability for low viscosities and Fig. 4 illustrates this behavior. It is shown that the shear mode is responsible of the numerical instability of the MRT-RM at low viscosities. In Fig. 5 and using the von Neumann stability condition, $|\lambda| \leq 1$, one can observe that the MRT-RM is unstable for low kinematic viscosity values, $\nu = \mu_\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$. Note that the parameters related with kinematic and bulk viscosity have been modified with respect to

Eq. (16), $s_4 = s_5 = s_6 = 1.994$ for $\nu = \mu_\nu = 5 \cdot 10^{-4} \frac{\text{kg}}{\text{ms}}$ and $s_4 = s_5 = s_6 = 1.988$ for $\nu = \mu_\nu = 1 \cdot 10^{-4} \frac{\text{kg}}{\text{ms}}$

Bulk viscosity stability:

One strategy to increase the stability of the LBM, successfully used by Lallemand and Luo [7], is to increase the value of the bulk viscosity of the fluid. In particular, they fixed the bulk viscosity value to $3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$). Figs. 6 and 7 compare the BGK, MRT-RM and MRT-CM with Lallemand's parameters:

$$S = \text{diag}(0, 0, 0, 1.64, 1.988, 1.988, 1.9, 1.9, 1.54). \quad (17)$$

Fig. 6(a) shows that variations on the bulk viscosity do not affect the dispersion values. However, Fig. 6(b) shows that increasing the bulk viscosity enhances dissipation. Comparing these results to Fig. 3(b) where viscosities were equal, we observe that only theoretical and numerical acoustic modes are affected.

Fig. 7 compares the numerical stability, which is controlled by the highest value of the dissipation mode for each wavenumber. It can be seen that the high dissipation of the acoustic modes originated from the high bulk viscosity proposed by Lallemand increases the stability of the MRT-RM with respect to the BGK. The stability of the MRT-CM with Lallemand's parameters is higher, because it is bounded by the shear mode due to the extra dissipation of the acoustic modes (see Fig. 6(b)).

In this section and using a von Neumann analysis, we have shown the effect of kinematic and bulk viscosities on the dispersion and dissipation errors. In particular, we can summarize the following conclusions:

- The MRT schemes add extra dissipation to the kinetic modes compared with the BGK.
- Viscosity variations do not affect dispersion errors.
- Bulk viscosity variations can improve stability through acoustic mode dissipation.
- Mach number variations (related to streaming velocity) modify the dissipation of shear and acoustic modes.
- The MRT-CM is more stable at low viscosities than MRT-RM. This MRT-RM instability comes from the shear mode.

The following section takes into account the above conclusions to propose new optimized free parameters s_6 , s_7 and s_8 that enhance the stability of the method, without damaging its accuracy.

4. Optimization of free parameters

As previously mentioned, the MRT collision operator relaxes each hydrodynamic moment independently, with different relaxation times. In D2Q9, the relaxation times s_0 , s_1 and s_2 are set to zero as it was mentioned in Section 2.2.2. Kinematic and bulk viscosities are controlled by s_3 , s_4 and s_5 while s_6 , s_7 and s_8 are free parameters. It should be noticed that these three degrees of

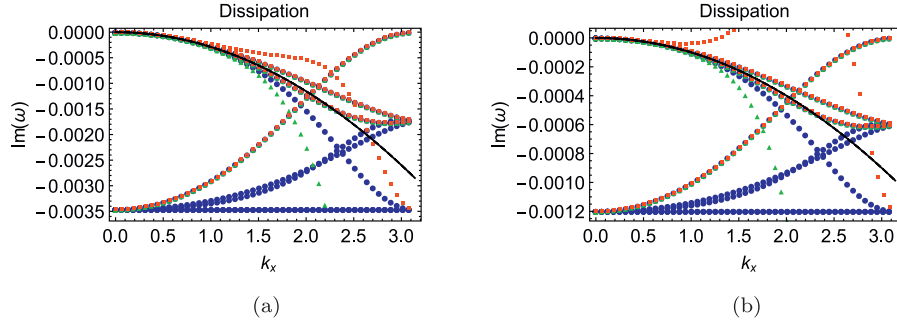


Fig. 4. Dissipation for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators and theoretical modes (shear mode, —, and acoustic mode, -) with $\nu = \mu_\nu = 5 \cdot 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.994$) (a) and $\nu = \mu_\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.988$) (b) at $\text{Ma} = 0.2$ (Note that the theoretical modes are overlapped).

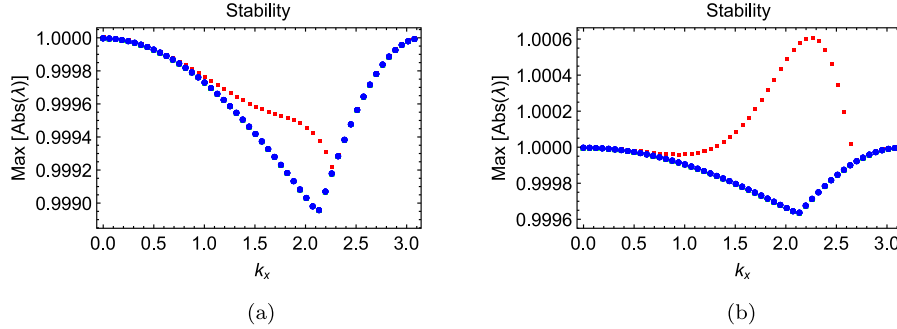


Fig. 5. Stability for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators with $\nu = \mu_\nu = 5 \cdot 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.994$) (a) and $\nu = \mu_\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.988$) (b) at $\text{Ma} = 0.2$ (Note that BGK and MRT-CM stabilities are overlapped).

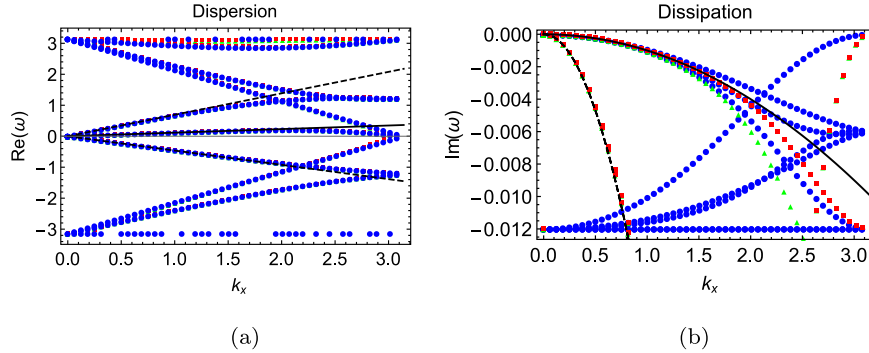


Fig. 6. Dispersion (a) and dissipation (b) for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators and theoretical modes (shear mode, —, and acoustic mode, -) with $\nu = 10^{-3} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.988$) and $\mu_\nu = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$) at $\text{Ma} = 0.2$.

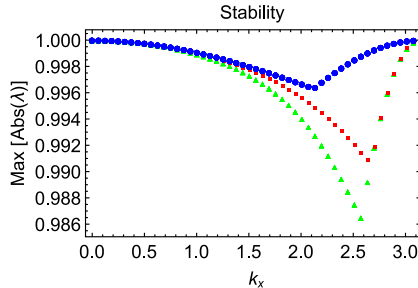


Fig. 7. Stability for BGK (●), MRT-RM (■) and MRT-CM (▲) collision operators with $\nu = 10^{-3} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.988$) and $\mu_\nu = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$) at $\text{Ma} = 0.2$.

freedom are usually reduced to two by setting $s_6 = s_7$ (to keep the symmetry in the moment relaxation).

The existence of these free parameters has motivated the optimization of the MRT. The first step in any optimization process is to define the objective of the optimization, and the information provided by von Neumann analysis (dispersion and dissipation er-

rors) is an interesting starting point. For example, the optimization of Lallemand and Luo [7] targeted the dispersion and dissipation errors while keeping in mind a minimization of anisotropy (to improve the lack of Galilean invariance of the model). Similarly, Xu and Sagaut [18], and Xu et al. [19] proposed a different optimization strategy to minimize dispersion/dissipation errors based on a minimization of the dissipation error. In this work a different approach has been followed where, without forgetting about dissipation and dispersion errors, the stability of the method has been the target of the optimization.

4.1. Rule of k -1% dispersion-error

The optimization described here has been inspired by the rule of k -1% dispersion-error proposed by Moura et al. [20] in the context of high order methods [27]. Moura et al. suggested that wavenumbers with dispersion errors higher than 1% should be dissipated, as high dispersion errors tend to pollute the solution. That means that a high dissipative error at high wavenumbers is a positive feature of a numerical method. Taking into account the

Table 1
k for the 1% dispersion error between numerical and theoretical mode in dispersion of the hydrodynamic modes.

Mode	$\nu = \mu_\nu$	BGK		MRT-RM		MRT-CM	
		Ma = 0	Ma = 0.2	Ma = 0	Ma = 0.2	Ma = 0	Ma = 0.2
ω^s	10^{-3}	1.107	1.217	1.107	1.217	1.107	1.217
	10^{-4}	1.107	1.217	1.107	1.217	1.107	1.217
	10^{-5}	1.107	1.217	1.107	1.217	1.107	1.217
ω^+	10^{-3}	0.584	0.526	0.605	0.542	0.605	0.542
	10^{-4}	0.584	0.526	0.605	0.542	0.605	0.542
	10^{-5}	0.584	0.526	0.605	0.542	0.605	0.542
ω^-	10^{-3}	0.584	0.856	0.605	0.887	0.605	0.903
	10^{-4}	0.584	0.856	0.605	0.887	0.605	0.903
	10^{-5}	0.584	0.856	0.605	0.887	0.605	0.903

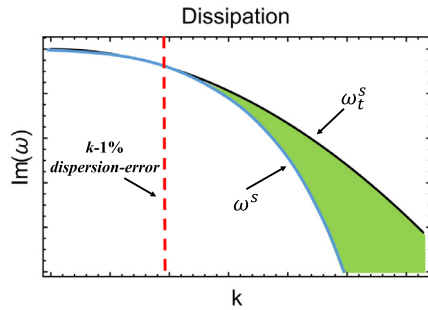


Fig. 8. Dissipation area calculated from the difference between the theoretical and numerical shear mode.

results of the previous section, we propose to optimize the values of the relaxation times s_6 , s_7 and s_8 to maximize the dissipation of the shear mode in wavenumbers higher than the k–1% dispersion-error wavenumber for the MRT–CM. The dissipation of acoustic modes has not been taken into account because, as it will be shown later, these barely change with the free parameters.

Table 1 shows the wavenumber where the error between the numerical and theoretical dispersion of hydrodynamic modes is 1%. The wavenumber values, k , have been obtained for two different Mach numbers (0 and 0.2) and three viscosities 10^{-3} , 10^{-4} and $10^{-5} \frac{\text{kg}}{\text{ms}}$ where $\nu = \mu_\nu$ ($s_{3,4,5} = 1.988, 1.9988$ and 1.99988). It must be remarked how the wavenumber values depend on the Mach number but show no dependency with the viscosity.

4.2. Results of the optimization process

This section details the optimization process. As explained before, the target of the optimization performed here is the maximization of the stability of the method. Taken into account the results obtained in Section 3.2, the maximization of the stability of the method is achieved by a maximization of the shear mode dissipation for wavenumbers, k , with a dispersion error larger than 1% (see Fig. 8).

The optimal values of s_{6-7} and s_8 (recall that $s_6 = s_7$ due to symmetry constraints) are found for different values of Mach number (Ma=0.1, 0.2 and 0.3) and viscosity ($\nu = \mu_\nu = 10^{-3}$, 10^{-4} and 10^{-5}). The parametric space (s_6 , s_8) is discretized with $\Delta s_i = 0.01$ (i.e. a tolerance of 1%). Besides, only values of the relaxation times in the range [0,2] are considered as defined in Section 2.2.2.

Table 2 shows the optimal values of these parameters. We only show values of s_6 and s_7 because s_8 does not affect the shear mode dissipation as illustrated in Fig. 9. As a result, s_8 has been fixed to 1.54, the value proposed by Lallemand and Luo [7]. Again, it can be observed that the values are dependent on Ma and are almost independent of the viscosity (see Table 2).

Fig. 10 shows the effect of the optimized free parameters in the dispersion and dissipation behavior. In Fig. 10(a), it is shown that

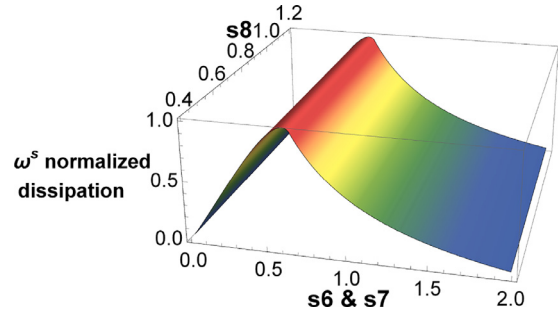


Fig. 9. Free parameters optimization based on shear mode dissipation surface at Ma = 0.1.

the dispersion of the hydrodynamic modes does not change when varying the free parameters. In this case, the k–1% dispersion-error for the optimized parameter at Ma = 0.2 are $\omega^s=1.233$, $\omega^+=0.542$ and $\omega^-=0.903$. These may be compared to the values in Table 1 to observe that they are almost identical. However, in Fig. 10(b) the shear mode shows higher dissipation for wavenumbers higher than the k–1% dispersion-error compared with Lallemand's parameters, as expected. Besides, the acoustic modes have the same behavior in both configurations producing the same numerical dissipation at high wavenumbers.

The numerical stability is shown in Fig. 11. We observe that this new result is identical (Fig. 11(a)) or better than Lallemand's parameters, when the bulk viscosity was increased (Fig. 11(b)), because in this case the stability is constrained by the shear mode instead of the acoustic mode.

For completeness, a two-dimensional dissipation map for the whole range of wavenumbers $\mathbf{k} = (k_x, k_y) \in [0 : \pi]$ is shown in Fig. 12. The maximum value of $\text{Im}(\omega)$ is plotted in order to quantify the dissipation of the different schemes. Additionally, unstable areas ($\text{Max}[\text{Im}(\omega)] > 0$) are marked with solid lines. Fig. 12(a) shows two areas where the MRT–RM with Lallemand's parameters becomes unstable at moderate wavenumbers as Malaspina showed in his work (see Fig. 3 in [13]). The MRT–CM with Lallemand's parameters remains stable for all combinations of wavenumbers $\mathbf{k}$ as shown in Fig. 12(b). Fig. 12(c) shows the effect of the optimized parameters where higher dissipation is observed at high wavenumbers. It should be mentioned that the MRT–CM with optimized parameter has a similar dissipation map than the one produced by Malaspina using a regularization BGK based on the expansion of Hermite polynomials. More demanding fluid conditions and further comparison for higher order equilibrium functions are included in Appendix C.

The optimization strategy, obtained in this section, has improved shear dissipation at high wavenumber, increasing the stability of the scheme. So far, we have proved that the proposed optimized scheme shows enhanced stability, but we still need to con-

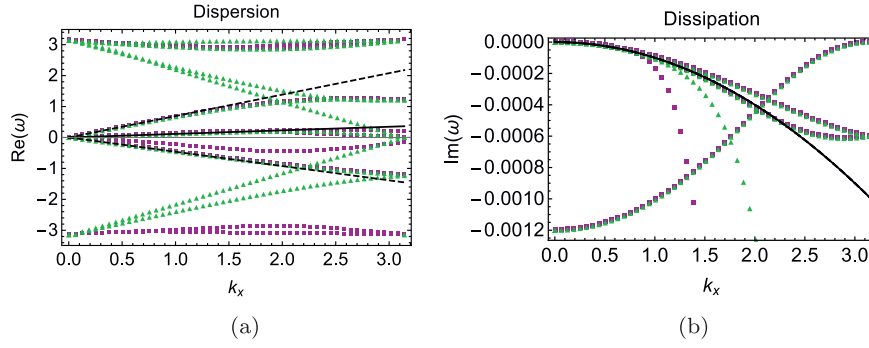


Fig. 10. Dispersion (a) and dissipation (b) with MRT-CM Lallemand ($\blacktriangle$) and optimized ($\blacksquare$); theoretical modes (shear mode, $-$, and acoustic mode, $-$) at $\nu = \mu_\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.9988$) and $\text{Ma} = 0.2$.

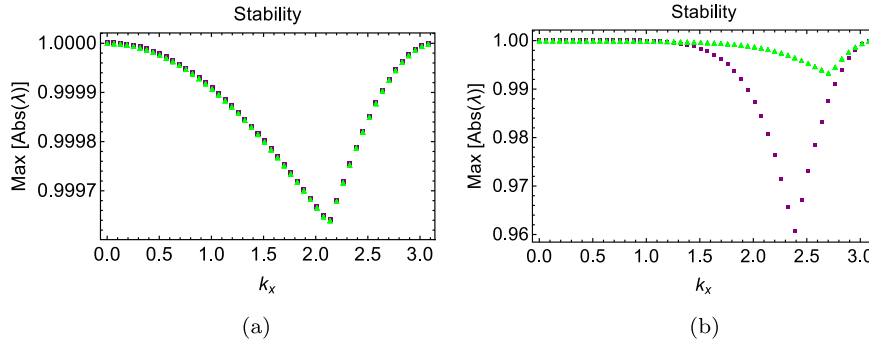


Fig. 11. Stability with MRT-CM Lallemand ($\blacktriangle$) and optimized ($\blacksquare$) parameters from k-1% dispersion-error with $\nu = \mu_\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{3,4,5} = 1.998$), (a) and with $\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.9988$) and $\mu_\nu = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$) (b) at $\text{Ma} = 0.2$.

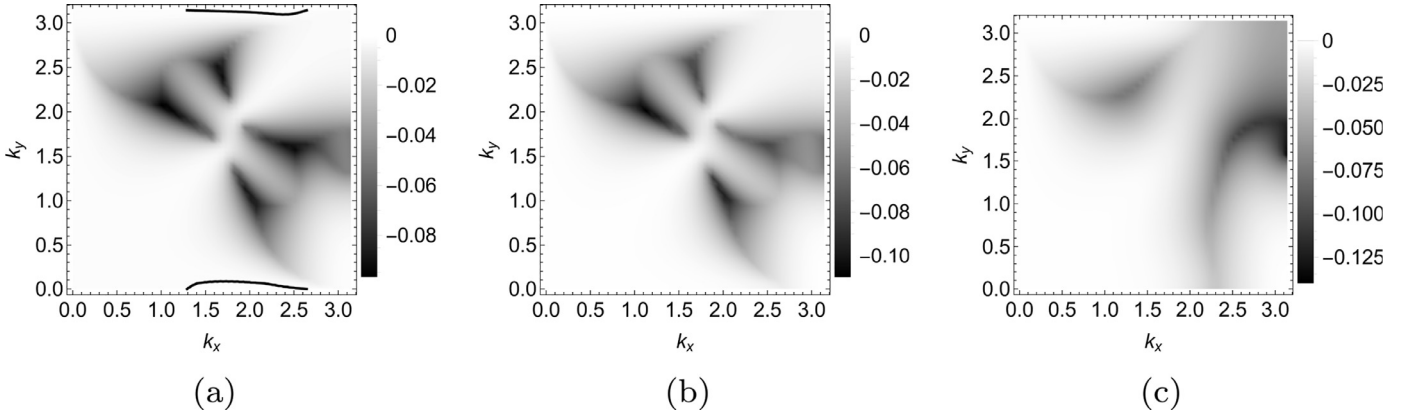


Fig. 12. Maximal value of the dissipation, $\text{Max}[\text{Im}(\omega)]$, for the MRT-RM (a), MRT-CM Lallemand (b) and optimized (c) parameters with $\nu = 10^{-4} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.9988$) and $\mu_\nu = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$) at $\text{Ma} = 0.2$.

firm that the accuracy is maintained. We will explore this point in the next section.

5. Numerical experiments

In this section the new optimized scheme is compared with previous approaches found in the literature. The aim of this section is to show that the proposed scheme provides increased stability without penalizing the accuracy.

The test case chosen for validation is a two dimensional Kelvin-Helmholtz instability. In this validation case an initial simple shear layer rolls-up with time due the instability excited by a $\mathcal{O}(\delta)$ perturbation in the vertical u_y velocity component. Minion and Brown [28] found that conventional numerical schemes (e.g. using centred differences) artificially create two spurious vortices if not enough

resolution is used. These spurious vortices result in a numerical instability for under-resolved simulations.

The boundary conditions for the simulation are periodic and the initial conditions for the perturbed shear layer read:

$$\begin{aligned} \rho &= \rho_0 \\ u_x &= \begin{cases} U_0 \tanh[w(y/L - 1/4)], & y \leq L/2 \\ U_0 \tanh[w(3/4 - y/L)], & y > L/2 \end{cases} \\ u_y &= U_0 \delta \sin[2\pi(x/L + 1/4)] \end{aligned} \quad (18)$$

where ρ_0 is the reference density fixed to 1. L is the number of grid points in each direction x and y , which defines a square domain. The parameters w and δ are related to the initial shear layer width and the magnitude of the initial perturbation; these are set to $w = 80$ and $\delta = 0.05$. The reference velocity, U_0 , is selected through the Mach number $\text{Ma} = U_0/c_s = 0.2$. The kinematic

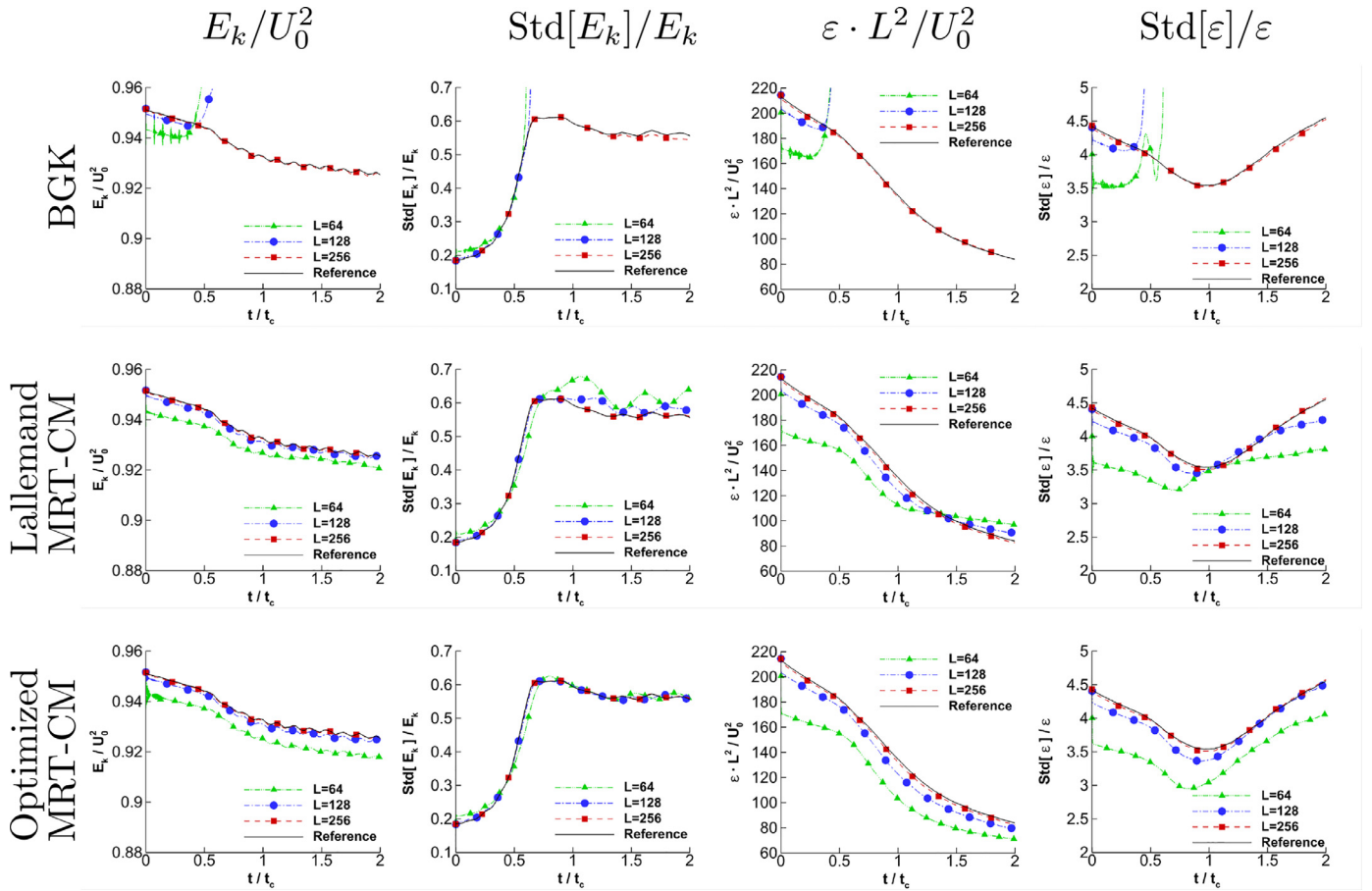


Fig. 13. Dimensionless mean kinetic energy, standard deviation of the kinetic energy, mean enstrophy and standard deviation of the enstrophy for various grid refinements ($L = 64$: $\blacktriangle$, $L = 128$: $\bullet$, $L = 256$: $\blacksquare$ and Reference: $-$), and collision operators (BGK: top row, MRT-CM Lallemand: middle row and MRT-CM Optimized: bottom row) with $Ma = 0.2$ and $Re = 3 \cdot 10^4$ at $t/t_c = 1$. All quantities are spatially averaged over the simulation domain.

viscosity is set through the relation with the Reynolds number, $Re = U_0 L / \nu = 3 \cdot 10^4$.

Dellar [29] recommended to set the bulk viscosity to 10 times the kinematic viscosity to prevent the apparition of spurious vortices. In this work we follow this recommendation and use a high bulk viscosity for all numerical schemes. Note that the bulk viscosity, μ_v , does not affect the optimization based on the shear mode dissipation (see Figs. 3(b) and 6(b)). Furthermore, when using higher bulk viscosities with the optimized parameters, the stability improves, as shown in Fig. 11(b).

Fig. 13 depicts the dimensionless mean kinetic energy, standard deviation of the kinetic energy, mean enstrophy and standard deviation of the enstrophy for different grid refinements ($L = 64$, 128, 256 and 512) and the three collision operators. The three chosen collision operators are the classical BGK, the MRT-CM with Lallemand and Luo parameters [7] and the MRT-CM with the optimized parameters derived in our work. Table 3 details the parameters used to produce Figs. 13 and 14. Note that s_4 and s_5 are defined by the viscosity that satisfies the Reynolds number.

It can be seen that for well resolved simulations (e.g. $L = 256$), the simulations converge to the reference solution and that the result obtained by the three collision operators are comparable. For coarser resolutions, the BGK tends to become unstable. Whilst both the MRT-CM with Lallemand's parameters and our approach remains stable, we observe that our optimization provides more consistent trends. Particularly, the advantages of the optimized parameters are shown in the standard deviation of the kinetic energy evolution, where it is reproduced correctly even for $L = 64$, where

the model with Lallemand's parameters provides an incorrect evolution. The mean enstrophy and fluctuations over time show similar behaviour, where the optimized MRT-CM for under-resolved resolutions provide a correct evolution tendency. The improvement in the stability of the method is similar to the one reported by Coreixas et al. [15] with a fourth-order recursive regularization method.

The discrepancies observed for under-resolved simulations are attributed to spurious vortices that pollute the simulations. Fig. 14 depicts the vorticity isocontours at $t/t_c = 1$ (where the characteristic time t_c is define as $t_c = L/U_0$) and shows spurious vortices for Lallemand's MRT-CM and BGK collision operators. However, we observe that our optimized MRT-CM approach is more stable and does not produce spurious vortices, even for under-resolved simulations.

In Section 4, we claimed that the optimization process does not affect the accuracy of the scheme for well resolved wavenumbers. This statement is supported by the results shown in Table 4 where the error of the simulations with $L = 64$, 128 and 256 are summarized. These errors are computed as the difference between the simulation and a reference solution (BGK collision operator with a fine grid with $L = 512$). It can be seen that the optimized MRT-CM presents lower numerical errors for under-resolved simulations while converging to Lallemand's MRT-CM for well resolved simulations (see Fig. 15). These results show that our optimization enhances the stability and improves accuracy in under-resolved grids.

For completeness, Fig. 16 shows the vorticity isocountours for MRT-CM collision operator and different values of $s_6 = s_7$ (the

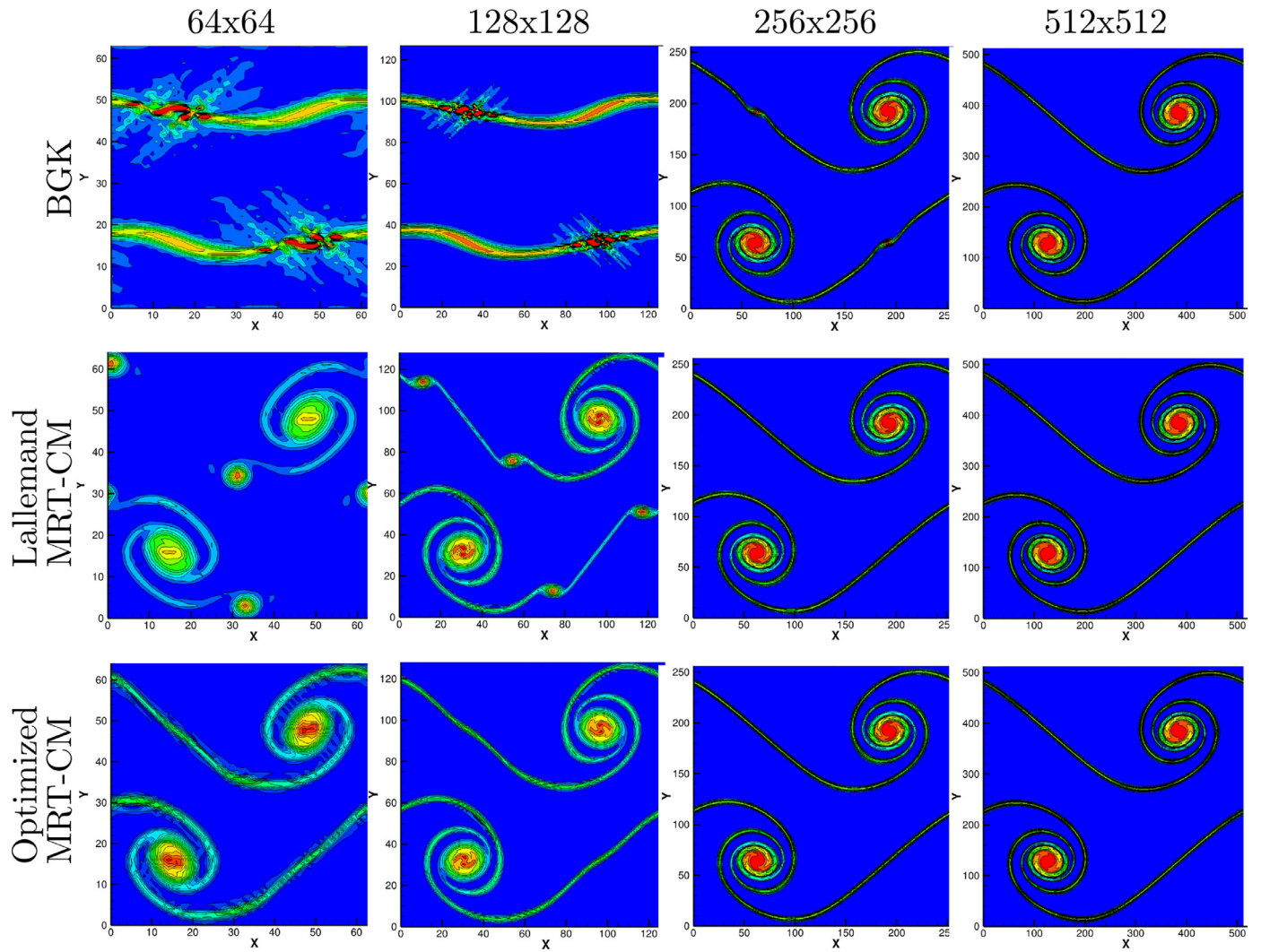


Fig. 14. Vorticity isocountours for different grid refinement and collision operators with $Ma = 0.2$ and $Re = 3 \cdot 10^4$ at $t/t_c = 1$.

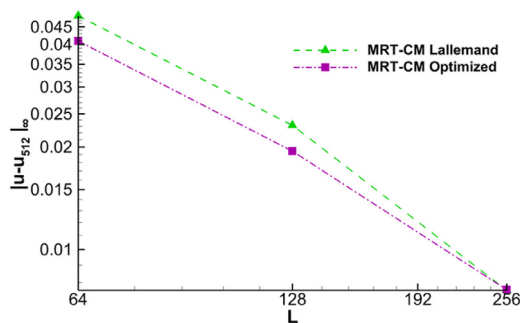


Fig. 15. Numerical Error $|u - u_{512}|_{\infty}$ for two MRT-CM collision operators (Lallemand vs optimized) with three resolutions: $L = 64, 128$ and 256 at $Ma = 0.2$ and $Re = 3 \cdot 10^4$ at $t/t_c = 1$.

Table 2
Optimized values for $s_6 = s_7$ at different viscosities and Ma number.

$\nu = \mu_\nu$	MRT-CM		
	$Ma = 0.1$	$Ma = 0.2$	$Ma = 0.3$
10^{-3}	0.63	0.94	1.13
10^{-4}	0.61	0.90	1.10
10^{-5}	0.61	0.90	1.10

Table 3

Relaxation times for the different collision operators.

BGK	Relaxation times $1/\tau$
MRT _{Lallemand}	(0, 0, 0, 1.64, s_4 , s_5 , 1.90, 1.90, 1.54)
MRT _{Optimized}	(0, 0, 0, 1.64, s_4 , s_5 , 0.90, 0.90, 1.54)

Table 4

Numerical Error $|u - u_{512}|_{\infty}$ for different collision operator and grid resolution at $Ma=0.2$ and $Re=3 \cdot 10^4$ at $t/t_c = 1$.

N	BGK	MRT-CM _{Lallemand}	MRT-CM _{Optimized}
64	–	4.79e–2	4.02e–2
128	–	2.28e–2	1.91e–2
256	8.75e–2	7.53e–3	7.59e–3

rest of the relaxation parameters are set to the values included in Table 2) with $L = 64$ at $t/t_c=1$. Fig. 16 supports the results obtained in Section 4, where the best solution (the one with diminished spurious vortices) is found for the optimized values (Table 2), which depend on the Mach number. Note that the optimized parameter are the ones found in the diagonal (from top left to bottom right).

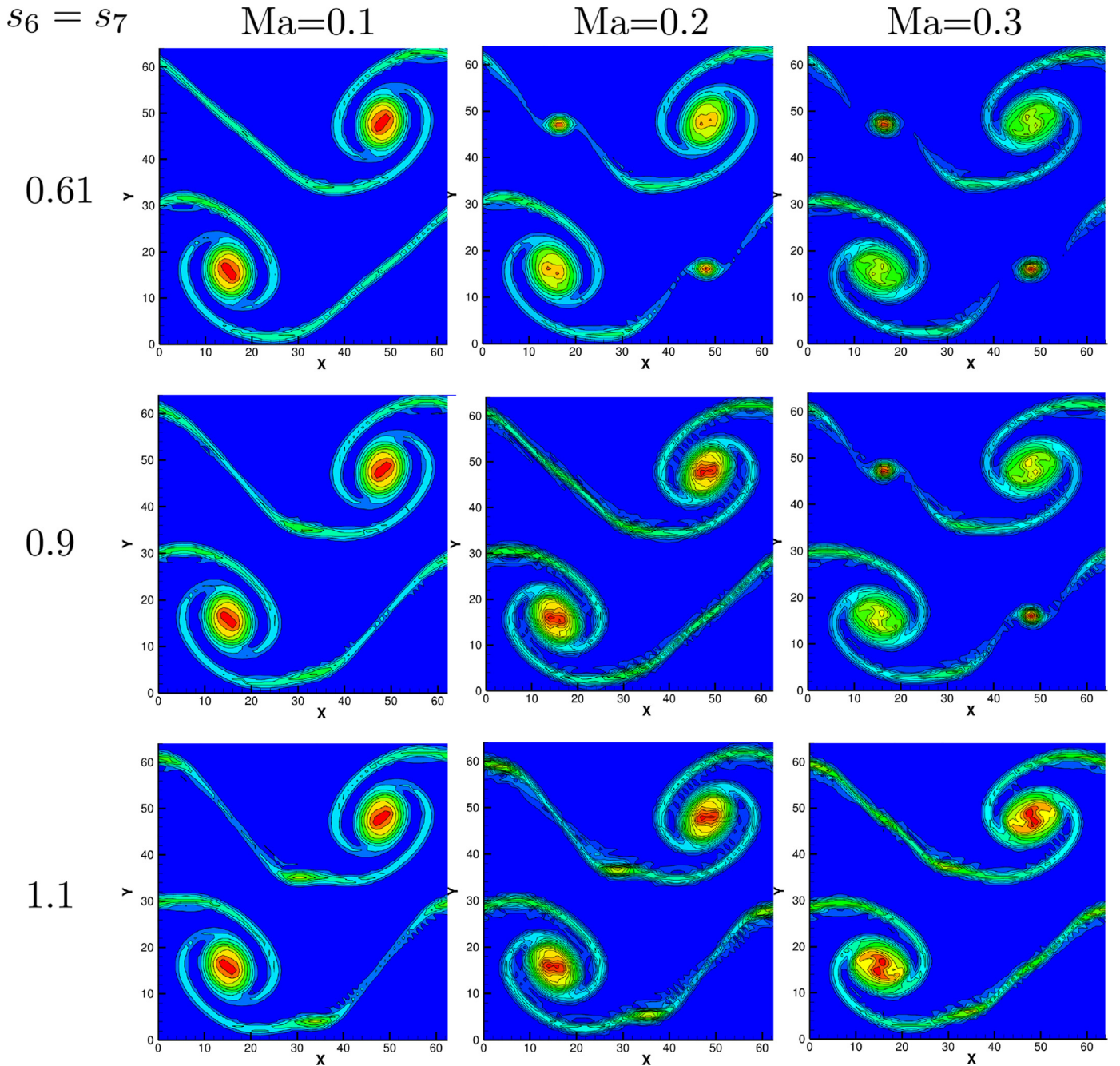


Fig. 16. Vorticity isocountours for different Ma and free parameters (from Table 2) at $Re = 3 \cdot 10^4$ with $L=64$, at $t/t_c = 1$.

6. Conclusions

This work proposes a new version of the multiple-relaxation lattice Boltzmann method with increased stability.

We use von Neumann analysis to analyze dispersion and dissipation errors in the lattice Boltzmann schemes including single (BGK) and multiple-relaxation times (MRT) for D2Q9 discretization. After showing the characteristics of the methods and illustrating the improved stability of the MRT over BGK schemes, we select the central moment MRT variant and optimize the two free parameters in the scheme to enhance its stability while retaining accuracy.

We optimize the parameters to obtain a scheme that dissipates flow features above the $k=1\%$ dispersion-error. This rule distinguishes between well resolved waves and under-resolved waves, by finding the wavenumber for which the dispersion error exceeds

a 1% error. By doing so, we can dissipate under-resolved features whilst maintaining accuracy which is dictated by lower wavenumbers.

We address the effects of grid resolution, varying laminar viscosities and streaming velocities (or Mach numbers) to show that the proposed optimized MRT-CM method is more stable and more accurate than previous schemes.

Acknowledgments

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Appendix A. Moments matrix for D2Q9

The $M(\mathbf{u})$ matrix is constructed by applying a Gram-Schmidt orthogonalization to the monomials $(\mathbf{e}_i - \mathbf{u})^n$ ordered in ascending order. The difference between the microscopic velocity and the reference velocity will be represented by $X = e_x - \tilde{u}$ and $Y = e_y - \tilde{v}$ to alleviate the notation. X and Y are vectors of dimension Q , and the operation $X^m Y^l$ is defined as a Hadamard product, therefore resulting in another vector of dimension Q .

Geier et al. [8] proposed the following set of moments:

$$X^m Y^l = (X^0 Y^0, X, Y, X^2 + Y^2, X^2 - Y^2, XY, X^2 Y, XY^2, X^2 Y^2)^T; \quad m, l = 0, 1, 2. \quad (\text{A.1})$$

Applying the Gram-Schmidt orthogonalization procedure to Eq. (A.1), the matrix $M(\mathbf{u})$ is obtained:

$$M(\mathbf{u}) = \begin{pmatrix} X^0 Y^0 \\ X \\ Y \\ 3(X^2 + Y^2) - 4 \\ X^2 - Y^2 \\ XY \\ 3X(X^2 + Y^2) - 5 \\ 3Y(X^2 + Y^2) - 5 \\ (9/2)(X^2 + Y^2)^2 - (21/2)(X^2 + Y^2) + 4 \end{pmatrix} = \begin{pmatrix} \rho | i \\ j_x | i \\ j_y | i \\ e | i \\ p_{xx} | i \\ p_{xy} | i \\ q_x | i \\ q_y | i \\ \epsilon | i \end{pmatrix} \quad (\text{A.2})$$

As can be seen, the selection of moments of Geier has a physical foundation. ρ is the density, e is the kinetic energy, ϵ is related to the kinetic energy square, $j_x = \rho u$ and $j_y = \rho v$ correspond to components of momentum, q_x and q_y correspond to the internal energy components and p_{xx} and p_{xy} correspond to the symmetric traceless viscous stress tensors.

Hence the transformation matrix in raw moments is $M(\mathbf{u} = 0)$ with $X = e_x$ and $Y = e_y$.

$$M(0) = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \end{pmatrix} \quad (\text{A.3})$$

Appendix B. Development of von Neumann stability analysis for LBM

The procedure of von Neumann analysis applied to LBM starts taking the PDF and splitting it into an equilibrium state, $\tilde{f}_i$, plus a perturbation, $\delta f_i(x, t)$:

$$f_i(x, t) = \tilde{f}_i + \delta f_i(\mathbf{x}, t). \quad (\text{B.1})$$

$\tilde{f}_i$ is the global PDF, which does not vary with time and space i.e., it depends only on the average density and velocity. The second term in the right hand side $\delta f_i(x, t)$ accounts for the fluctuations from equilibrium.

The first step consist in substituting Eq. (B.1) into Eq. (3) and expanding the f_i^{eq} by means of Taylor series centred at the global distribution, $\tilde{f}_i$ [7]:

$$\Omega_j = K_{ji} \left[\left(\tilde{f}_i + \frac{\partial f_i^{eq}}{\partial f_j} \delta f_j^{eq}(\mathbf{x}, t) + O((\delta f_i)^2(\mathbf{x}, t)) \right) - (\tilde{f}_i + \delta f_i(\mathbf{x}, t)) \right]. \quad (\text{B.2})$$

Additionally, the partial derivate of the collision operator can be expressed as:

$$\frac{\partial f_i^{eq}(\rho, u)}{\partial f_j} = \left(\frac{\partial f_i^{eq}}{\partial \rho} \frac{\partial \rho}{\partial f_j} + \frac{\partial f_i^{eq}}{\partial u} \frac{\partial u}{\partial f_j} \right). \quad (\text{B.3})$$

Substituting Eq. (B.1) into Eq. (2):

$$\delta f_i(\mathbf{x} + \Delta \mathbf{x}, t + \Delta t) - \delta f_i(\mathbf{x}, t) = \Delta t K_{ji} \left(\frac{\partial f_i^{eq}}{\partial f_j} \delta f_j(\mathbf{x}, t) - \delta f_i(\mathbf{x}, t) \right). \quad (\text{B.4})$$

Once the collision operation Eq. (B.2) has been linearized, the solution has been assumed as:

$$\delta f_i(\mathbf{x}, t) = F_i e^{i(\mathbf{k}\mathbf{x} - \omega t)}, \quad (\text{B.5})$$

where F_i is the amplitude of a sinusoidal wave at the position $\mathbf{x}$ at a given time t , $k = \{k_x, k_y\}$ denotes the wavenumber and ω is the frequency in time.

Replacing Eq. (B.5) into (B.4) and gathering terms:

$$F_i e^{-i\omega \Delta t} = F_j e^{-i\mathbf{k}\Delta \mathbf{x}} \left[\delta_{ij} + \Delta t K_{ji} \left(\frac{\partial f_i^{eq}}{\partial f_j} - \delta_{ij} \right) \right], \quad (\text{B.6})$$

where δ_{ij} is the Kronecker delta function and $\Delta \mathbf{x} = \mathbf{e}_i \Delta t$. Identifying terms the amplification matrix G_{ij} for the collision operator reads:

$$G_{ij} = A_{ik}^{-1} [\delta_{kj} + \Delta t K_{kn} N_{nj}], \text{ with } \begin{cases} A_{ik} = \delta_{ik} e^{i\mathbf{k}\Delta t \mathbf{e}_i}, \\ N_{nj} = \frac{\partial f_j^{eq}}{\partial f_n} - \delta_{nj}. \end{cases} \quad (\text{B.7})$$

Appendix C. MRT-CM with fourth order f_i^{eq}

Let us now consider more demanding fluid conditions based on Malaspinas' work (see Fig. 4 in [13]): $\mu_\nu = 3.3 \cdot 10^{-5} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.9996$). Fig. C.17(a) shows that unstable modes appear for $\text{Ma} = 0.28$. The maximum Mach number before instabilities appear is lower than for Malaspinas' method, where the limit is reached for $\text{Ma} = 0.43$. However, in Malaspinas' approach a fourth order equilibrium function f_i^{eq} is used, while we only retained second order terms in our work. The effect of the f_i^{eq} fourth order in the MRT-CM to figure out the Mach limitation is shown in Fig. C.17.

To show how the expansion order of f_i^{eq} affects the maximum Mach number limitation, we compare the second (Eq. (4)) and (fourth order (Eq. (C.1)) extensions for f_i^{eq} .

The f_i^{eq} fourth order is defined as:

$$f_i^{eq} = \rho w_i \left(1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} - \frac{\mathbf{e}_i \cdot \mathbf{u}^3}{2c_s^5} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^3}{6c_s^6} + \frac{\mathbf{u}^4}{8c_s^4} - \frac{\mathbf{e}_i^2 \cdot \mathbf{u}^4}{4c_s^6} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^4}{24c_s^8} \right), \quad (\text{C.1})$$

Fig. C.17(b) shows that the unstable area visible for second order expansions (at $\text{Ma} = 0.28$) disappears when using a fourth order equilibrium function. For completeness, we increase the Mach number for the 4th order expansion until instabilities arise. We find that the limiting Mach is in this case $\text{Ma} = 0.43$, which is the same as found by Malaspinas. Fig. C.17(c) illustrates the dissipation map for this last case: MRT-CM with 4th order expansions for f_i^{eq} .

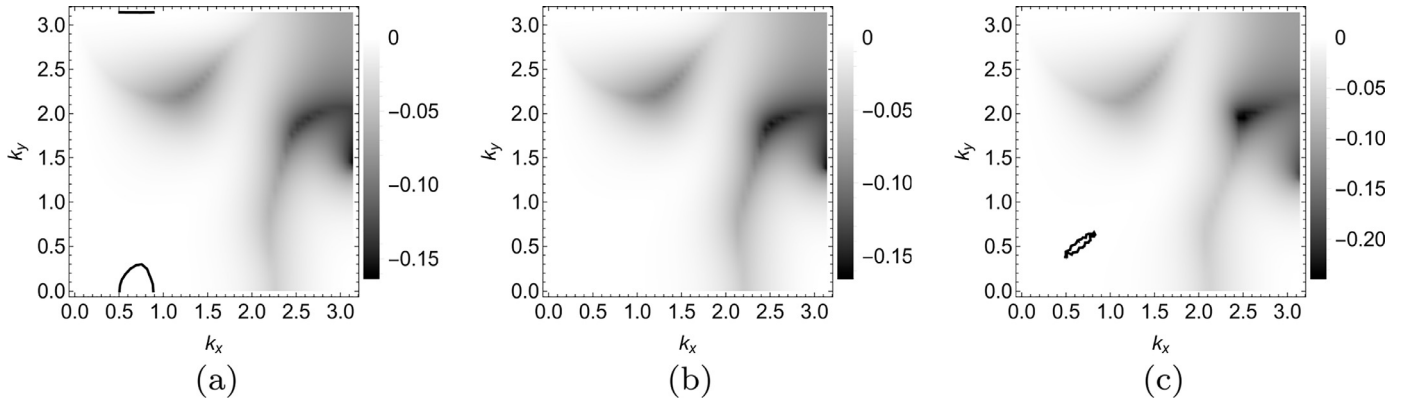


Fig. C17. Maximal value of the dissipation, $\text{Max}[\text{Im}(\omega)]$, for the MRT-CM optimized with 2nd order f_i^{eq} at $\text{Ma} = 0.28$ (a), 4th order f_i^{eq} at $\text{Ma} = 0.28$ (b) and 4th order f_i^{eq} at $\text{Ma} = 0.43$ (c). The viscosities set to: $\nu = 3.3 \cdot 10^{-1} \frac{\text{kg}}{\text{ms}}$ ($s_{4,5} = 1.9996$) and $\mu_v = 3.66 \cdot 10^{-2} \frac{\text{kg}}{\text{ms}}$ ($s_3 = 1.64$).

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